

Linear Approximation and L'Hôpital's Rule



Tangent line approximation

- 1 Use linear approximation to estimate $\sqrt{4.1}$, given $f(x) = \sqrt{x}$, $f(4) = 2$, and $f'(4) = \frac{1}{4}$.
- 2 Use linear approximation to estimate $(1.02)^5$, using $f(x) = x^5$ near $x = 1$.
- 3 Find the linearization $L(x)$ of $f(x) = \sin x$ at $x = 0$, and use it to approximate $\sin(0.1)$.

Error and over/underestimation

- 4 For $f(x) = \sqrt{x}$ (concave down for $x > 0$), explain whether the linear approximation at $x = 4$ overestimates or underestimates $f(4.1)$.
- 5 For $f(x) = x^2$ (concave up everywhere), explain whether the tangent line approximation at any point overestimates or underestimates nearby function values.

Indeterminate forms and L'Hôpital's Rule

- 6 Evaluate $\lim_{x \rightarrow 0} \frac{\sin x}{x}$ using L'Hôpital's Rule, and confirm it matches the known limit.
- 7 Evaluate $\lim_{x \rightarrow 0} \frac{e^x - 1 - x}{x^2}$ using L'Hôpital's Rule (applied twice).
- 8 Evaluate $\lim_{x \rightarrow \infty} \frac{\ln x}{x}$ using L'Hôpital's Rule.

Recognizing when L'Hôpital's Rule applies

- 9 Explain why L'Hôpital's Rule cannot be directly applied to $\lim_{x \rightarrow 0} \frac{x+1}{x}$, and evaluate the limit by other means.

Differentials

- 10 For $f(x) = x^3$, find the differential dy when $x = 2$ and $dx = 0.1$, and compare it to the actual change $\Delta y = f(2.1) - f(2)$.

More L'Hôpital's Rule practice

11 Evaluate $\lim_{x \rightarrow 0} \frac{\tan x - x}{x^3}$ using L'Hôpital's Rule (applied as many times as needed).